Chapter 7 Answers

Problem 1

Note: It should be noted that if a system is complete, then representability implies complete representability. For suppose that A is representable by h under some complete system $\mathcal{S}$. Thus $h(n) \in P$ iff $n \in A$. On the one hand, if $h'(n) \in P$ then $h(n) \in R$ by definition, and thus $n \notin A$, meaning $n \in \bar{A}$. If $n \in \bar{A}$, then $h(n) \notin P$, and because the system is complete, $h(n) \in R$ and thus $h'(n) \in P$ (if the system was not complete, it could be that $h(n)$ was undecidable). Thus h completely represents A under the complete system $\mathcal{S}$.

(a): On the one hand, suppose h defines A. Thus (1) $n \in A$ implies $h(n) \in P$, and (2) $n \in \bar{A}$ implies $h(n) \in R$ (which is the same as saying $h'(n) \in P$). We need to show that if $h(n) \in P$ implies $n \in A$ and $h(n) \in R$ implies $n \in \bar{A}$. Suppose $h(n) \in P$. Thus $h(n) \notin R$ meaning $n \in A$ by (2). If $h(n) \in R$, then $h(n) \notin P$ and thus $n \in \bar{A}$ by (1). Thus h completely represents A. On the other hand, suppose that h completely represents A. It is obvious that h then defines A.

(b): Suppose that h contra-represents A. Thus $h(n) \in R$ iff $n \in A$, which is the same as saying $h'(n) \in P$ iff $n \in A$, meaning h' represents A. If h' represents A, then $h'(n) \in A$ iff $n \in A$, which is the same as saying $h(n) \in R$ iff $n \in A$. Thus A is contra-representable by h.

Problem 2

(a): On the one hand, suppose that h represents A. Thus $h(n) \in P$ iff $n \in A$, which is the same as saying $r_h(n) \in P$ iff $n \in A$. Because $n \in r_h^{-1}(P)$ iff $f(n) \in P$, this means that $n \in r_h^{-1}(P)$ iff $n \in A$. On the other hand, suppose that $A = r_h^{-1}(P)$. Thus $n \in A$ iff $r_h(n) \in P$, which is the same as saying $n \in A$ iff $h(n) \in P$.

(b): Suppose that h contra-represents A. Thus $n \in r_h^{-1}(R)$ iff $h(n) \in R$ iff $n \in A$. On the other hand, suppose that $A = r_h^{-1}(R)$. Thus $h(n) \in R$ iff $n \in r_h^{-1}(R)$ iff $n \in A$.

Problem 3

Suppose that A is represented in $\mathcal{S}$ by h, while $\bar{A}$ is not represented by any number. As a consequence of the note in Problem 1, if $\mathcal{S}$ was complete, $\bar{A}$ would be represented by h'. Thus, because $\bar{A}$ is not represented by h', $\mathcal{S}$ is incomplete.

Problem 4

If $\bar{P^*}$ was representable in $\mathcal{S}$, then there would be some number h such that $h(n) \in P$ iff $n \in \bar{P^*}$. Thus $h(h) \in P$ iff $h \in \bar{P^*}$ iff $h(h) \notin P$, which is impossible. Now suppose that $\bar{R^*}$ was representable in $\mathcal{S}$ by h. Thus $h(n) \in P$ iff $n \in \bar{R^*}$, and $h'(h') \in R$ iff $h(h') \in P$ iff $h' \in \bar{R^*}$ iff $h'(h') \notin R$, which is impossible.

Problem 5

(a): Suppose that h represents R^*. Thus $h(n) \in P$ iff $n \in R^*$, meaning $h(h) \in P$ iff $h \in R^*$ iff $h(h) \in R$, which means that $h(h)$ is undecidable.

(b): Suppose that h represents P^*. Thus $h(n) \in P$ iff $n \in P^*$, meaning $h(h') \in P$ iff $h' \in P^*$ iff $h'(h') \in P$ iff $h(h') \in R$, which means that $h(h')$ is undecidable.

Problem 6

(a): Suppose that h represents some superset of R^* disjoint from P^*. Thus $h(n) \in P$ iff $n \notin P^*$, meaning $h(h) \in P$ iff $h \notin P^*$ iff $h(h) \notin P$, meaning $h(h)$ is undecidable.

(b): Suppose that h represents some superset of P^* disjoint from R^*.

Thus $h(n) \in P$ iff $n \notin R^*$, meaning $h(h') \in P$ iff $h' \notin R^*$ iff $h'(h') \notin R$ iff $h(h') \notin P$, meaning $h(h')$ is undecidable.

The reason that these results are stronger than those of Problem 5 is that the results of Problem 5 are special cases of Problem 6. To see this, note that P^* is disjoint from R^* for the following reason: R and P are defined to be disjoint, and $h \in P^*$ iff $h(h) \in P$ iff $h(h) \notin R$ iff $h \notin R^*$, thus proving that P^* and R^* are disjoint. Thus the results of Problem 5 are special cases of Problem six, taking the superset of R^* disjoint from P^* to be R^* itself, and the superset of P^* disjoint from R^* to be P^* itself.

Problem 7

Consider any sets A and B. Then:

(a): Suppose $A \subseteq B$. Now, $x \in f^{-1}(A)$ iff $f(x) \in A$, and $x \in f^{-1}(B)$ iff $f(x) \in B$. Thus $x \in f^{-1}(A)$ implies $x \in f^{-1}(B)$, meaning $f^{-1}(A) \subseteq f^{-1}(B)$.

(b): Suppose that A is disjoint from B. Because $x \in f^{-1}(A)$ iff $f(x) \in A$, and $x \in f^{-1}(B)$ iff $f(x) \in B$, x cannot be in both $f^{-1}(A)$ and $f^{-1}(B)$, or else $f(x)$ would be in both A and B.

(c): We know that $x \in f^{-1}(A)$ iff $f(x) \in A$ and $x \in f^{-1}(\bar{A})$ iff $f(x) \in \bar{A}$, thus meaning $x \in f^{-1}(A)$ iff $x \notin f^{-1}(\bar{A})$, therefore meaning they are complements.

Problem 8

Suppose that $f(x)$ is admissible. Then:

(a): Suppose that A is representable by h. Thus $h(n) \in P$ iff $n \in A$, and there is a number k such that for all for every n $k(n)$ is equivalent to $h(f(n))$. Now $x \in f^{-1}(A)$ iff $f(x) \in A$. Thus $k(n) \in P$ iff $h(f(n)) \in P$ iff $f(n) \in A$ iff $n \in f^{-1}(A)$, thus proving that $f^{-1}(A)$ is representable by $k(n)$. On the other hand, suppose that A is contra-representable by h. Thus $h(n) \in R$ iff $n \in A$. Hence $k(n) \in R$ iff $h(f(n)) \in R$ iff $f(n) \in A$ iff $n \in f^{-1}(A)$, thus proving that $f^{-1}(A)$ is contra-representable by $k(n)$.

(b): Consider some exactly separable disjoint pair (A, B) under some h in $\mathcal{S}$. Thus h represents A and contra-represents B. Taking $k(n)$ in the

same way as in part (a), $f^{-1}(A)$ is representable by $k(n)$, while $f^{-1}(B)$ is contra-representable by $k(n)$, thus meaning the pair $(f^{-1}(A), f^{-1}(B))$ is exactly separable in $\mathcal{S}$.

Problem 9

Suppose that $\mathcal{S}$ is a normal system, which means that for every number h, $h^{\#}(n)$ is equivalent to $h(d(n))$. Now suppose that A is representable by some h. Thus $h(n) \in P$ iff $n \in A$, and $h^{\#}(n) \in P$ iff $h(d(n)) \in P$ iff $d(n) \in A$. Thus $h \in A^*$ iff $h(h) \in A$ iff $h(d(h)) \in P$ iff $h^{\#}(h) \in P$, which means $h^{\#}$ represents A^* in $\mathcal{S}$.

Problem 10

Consider some normal system $\mathcal{S}$, meaning, for every h, $h^{\#}(n)$ is equivalent to $h(d(n))$. Now suppose that $\bar{P}$ was representable in $\mathcal{S}$ by h. Thus $h(n) \in P$ iff $n \in \bar{P}$. Thus $h^{\#}(h^{\#}) \in P$ iff $h(h^{\#}(h^{\#})) \in P$ iff $h^{\#}(h^{\#}) \in \bar{P}$, which is impossible. On the other hand, suppose that $\bar{R}$ was representable by h. Thus $h(n) \in P$ iff $n \in \bar{R}$. Thus $h^{\#'}(h^{\#'}) \in R$ iff $h^{\#}(h^{\#'}) \in P$ iff $h(h^{\#'}(h^{\#'})) \in P$ iff $h^{\#'}(h^{\#'}) \in \bar{R}$, which is impossible.

We can also prove this more swiftly. By Problem 9, if $\bar{P}$ is representable, then so is $\bar{P}^*$, which is equal to $\bar{P}^*$ by Problem 7. But by Problem 4, $\bar{P}^*$ cannot be representable. Thus $\bar{P}$ cannot be representable. A similar argument shows that $\bar{R}$ is not representable.

Problem 11

Consider some normal system $\mathcal{S}$. Thus for every number h, $h^{\#}(n)$ is equivalent to $h(d(n))$. Thus $h(h^{\#}(h^{\#}))$ is equivalent to $h^{\#}(h^{\#})$, meaning $h^{\#}(h^{\#})$ is a fixed point for h.

Problem 12

(a): Suppose that h represents R, meaning $h(n) \in P$ iff $n \in R$. If n is a fixed point of h, then $h(n) \in P$ iff $n \in P$. But $h(n) \in P$ iff $n \in R$, meaning n must be undecidable.

(b): Suppose that h represents P, meaning $h(n) \in P$ iff $n \in P$. If n is a fixed point of h', then $n \in P$ iff $h(n) \in P$ iff $h'(n) \in R$ iff $n \in R$, which means that n is undecidable. Also, $h'(n) \in P$ iff $n \in P$ iff $h(n) \in P$, thus showing that $h'(n)$ (and $h(n)$) is also undecidable.

Exercise 1

I could not solve either of these.

Problem 13

Suppose that A is some superset of R^* disjoint from P^*. If it was definable by some h in $\mathcal{S}$, we would have (1) $n \in A$ implies $h(n) \in P$, and (2) $n \in \bar{A}$ implies $h(n) \in R$. Thus $h \in \bar{A}$ implies $h(h) \in R$ meaning $h \in R^*$. But becuse $h \in \bar{A}$, $h \notin R^*$, and thus $h \notin \bar{A}$, that is, $h \in A$. Now if $h \notin P^*$, then $h(h) \notin P$, which would imply by (1) that $h \notin A$, a contradiction. Thus $h \in P^*$, contradicting the fact that A was disjoint from P^*, meaning A cannot be definable in $\mathcal{S}$.

Exercise 2

Suppose that $P^* \subseteq A$, and h defines A in $\mathcal{S}$. Thus (1) $n \in A$ implies $h(n) \in P$, and $n \notin A$ implies $h(n) \in R$. Suppose that $h' \notin A$. Thus $h(h') \in R$ by (2), meaning $h'(h') \in P$ and $h' \in P^*$, and because $P^* \subseteq A$, $h' \in A$, which is a contradiction. Thus $h' \in A$. Now suppose that $h' \notin R^*$. Thus $h'(h') \notin R$ meaning $h(h') \notin P$, which would mean that $h' \notin A$ by (1), which is a contradiction. Thus $h' \in R^*$.

Problem 14

Note that for all of the following problems, we assume that $\mathcal{S}$ is a standard simple system.

Consider some standard simple system $\mathcal{S}$ and any recursive function $f(x)$. Then:

(1') Suppose A is recursively enumerable. Now, $f^{-1}(A)$ is the set of all x such that $f(x) \in A$. Thus $x \in f^{-1}(A)$ iff $\exists y(f(x) = y \wedge y \in A)$, which is recursively enumerable.

(2') Suppose that A is recursive. Thus by part (1'), both $f^{-1}(A)$ and $f^{-1}(\bar{A})$ are recursively enumerable. Because $f^{-1}(A)$ and $f^{-1}(\bar{A})$ are complements by Problem 7(c), it follows that $f^{-1}(A)$ is recursive.

Note that (1) and (2) of this problem follow by taking $f(x)$ to be $r(x, x)$ $(d(x))$, which is recursive as $\mathcal{S}$ is a standard system.

Problem 15

(a): Suppose that A is representable in $\mathcal{S}$ by h. Thus, for all n, $r(h, n) \in P$ iff $n \in A$. Because $r(h, n) \in P$ is recursively enumerable, so is A.

(b): Suppose that A is definable in $\mathcal{S}$ by h. Thus (1) $n \in A$ implies $r(h, n) \in P$ and (2) $n \in \bar{A}$ implies $r(h, n) \in R$. These imply that $r(h, n) \in P$ implies $n \in A$, and $r(h, n) \in R$ implies $n \in \bar{A}$. Thus $r(h, n) \in P$ and $r(h, n) \in R$ represent A and $\bar{A}$, respectively. By part (a) of this problem, both A and $\bar{A}$ are thus recursively enumerable, meaning A is recursive.

Problem 16

(a): Suppose that some recursively enumerable but non-recursive set A is representable in $\mathcal{S}$. There are two cases: Either there is some number n outside of A, or there isn't. Suppose that there was some n outside of A. By Problem 15, we know that A cannot be definable in $\mathcal{S}$, or else A would be recursive. Hence, because condition (1) of being definable already holds because A is representable, condition (2) must not hold (for the h which represents A). Thus $n \in \bar{A}$ does not imply $r(h, n) \in R$. Now, we know that there is some $n \in \bar{A}$. Because $r(h, n) \notin R$, it must be undecidable, for being in P would imply

that $n \in A$ (because h represents A). Thus $r(h, n)$ is undecidable for any n outside of A. On the other hand, suppose that there were no n outside of A. But this is impossible, as that would mean $\bar{A}$ is the empty set, a recursively enumerable set, contrary to the assumption that A was not recursive.

A more swift proof can be seen in this way: Because A is not recursive, $\bar{A}$ is not recursively enumerable, and hence not representable (because all representable sets in $\mathcal{S}$ are recursively enumerable by Problem 15). Thus, because A is represented in $\mathcal{S}$, but not $\bar{A}$, $\mathcal{S}$ is incomplete by Problem 3.

(b): Suppose that all recursively enumerable sets are representable in $\mathcal{S}$. Then there must be some recursively enumerable non-recursive set representable in $\mathcal{S}$. For example, the set T_0 of the Gödel numbers of the true sentences of the system (U) defined in chapter 4 is one such set. Hence, by part (a) of this problem, $\mathcal{S}$ is incomplete.

Problem 17

Suppose that $\mathcal{S}$ is undecidable and standard. Thus both P and R are recursively enumerable. So if there was not some number outside of both P and R, R would be the complement of P, meaning P would be recursive, a contradiction to the assumed undecidability. Hence there must be some number outside of both P and R, meaning $\mathcal{S}$ is incomplete.

Problem 18

Suppose that all recursive sets are representable in $\mathcal{S}$. Now, suppose that $\mathcal{S}$ was not undecidable, which would mean that P is recursive. By Problem 14(1), P^* is also recursive. Hence $\bar{P}^*$ is also recursive, and therefore representable in $\mathcal{S}$ by some h. Thus, for all n, $h(n) \in P$ iff $n \in \bar{P}^*$. This means $h(h) \notin P$ iff $h \in P^*$ iff $h(h) \in P$, which is impossible. Thus $\mathcal{S}$ being decidable raises a contradiction, therefore meaning it must be undecidable.

Problem 19

Suppose that the pair (P, R) is recursively inseparable. Hence there is no recursive superset of P disjoint from R. So if $\mathcal{S}$ was not undecidable, P

would be recursive. Because P is a superset of itself, this is a contradiction to the assumed recursive inseparability. Thus $\mathcal{S}$ must be undecidable.

Problem 20

Consider any recursive function $f(x)$ and any disjoint pair (A, B). Suppose $(f^{-1}(A), f^{-1}(B))$ is recursively inseparable. Hence there is no recursive superset of $f^{-1}(A)$ disjoint from $f^{-1}(B)$. Now suppose that (A, B) was not recursively inseparable. Hence some superset S of A disjoint from B is recursive. But $x \in f^{-1}(S)$ iff $f(x) \in S$. Thus $f^{-1}(S)$ is equivalent to $f(x) \in S$, meaning $f^{-1}(S)$ is recursive. Now, $f^{-1}(S)$ is a supserset of $f^{-1}(A)$, as $x \in f^{-1}(A)$ iff $f(x) \in A$, which would mean $f(x) \in S$ iff $x \in f^{-1}(S)$. Also, $f^{-1}(S)$ is disjoint from $f^{-1}(B)$, as $x \in f^{-1}(B)$ iff $f(x) \in B$, which would mean $f(x) \notin S$ iff $x \notin f^{-1}(S)$. Thus, because $f^{-1}(S)$ is a superset of $f^{-1}(A)$ disjoint from $f^{-1}(B)$, this is contrary to the assumption of recursive inseparability for $(f^{-1}(A), f^{-1}(B))$. Thus (A, B) must be recursively inseparable. Thus if (P^*, R^*) is recursively inseparable, so is (P, R), as $P^* = d^{-1}(P)$ and $R^* = d^{-1}(R)$.

Problem 21

Suppose that every recursive set is definable in $\mathcal{S}$. Suppose that (P^*, R^*) was not recursively inseparable. Hence there would be some recurisve superset of P^* disjoint from R^*, and because the superset is recursive, it would be definable in $\mathcal{S}$. But this is impossible by Problem 13. Thus (P^*, R^*) is recursively inseparable, meaning so is (P, R) by Problem 20.

Problem 22

Consider some standard simple system $\mathcal{S}$, and suppose that some disjoint pair (A_1, A_2) is recursively inseparable, but separable in $\mathcal{S}$. Thus some number h represents some superset S_1 of A_1 and contra-represents some superset S_2 of A_2. Suppose that S_1 and S_2 were not disjoint, that is, they shared some number n. Thus $h(n) \in P$ and $h(n) \in R$, which is impossible. Therefore S_1 and S_2 are disjoint. Because (A_1, A_2) is recursively inseparable, so is (S_1, S_2). For if it weren't, there would be some recursive superset S of S_1 disjoint from S_2. But S would also be a superset of A_1, and because it is disjoint from S_2, would

also be disjoint from A_2, in contradiction to the assumed recursive inseparability of (A_1, A_2). Thus (S_1, S_2) is recursively inseparable. Now, $S_1 = r_h^{-1}(P)$ and $S_2 = r_h^{-1}(R)$ by Problem 2. Thus the pair $(r_h^{-1}(P), r_h^{-1}(R))$ is recursively inseparable, meaning so is (P, R) by Problem 20.

Corollary 1 follows in this way: Suppose some recursively inseparable pair is separable in $\mathcal{S}$. By Theorem 4, (P, R) is recursively inseparable. Thus by Problem 19, $\mathcal{S}$ is undecidable.

Corollary 2 follows in this way: Suppose some recursively inseparable pair is separable in $\mathcal{S}$. Thus $\mathcal{S}$ is undecidable by Corollary 1, and is thus also incomplete by Theorem 1.

Problem 23

Note: It should be noted that the second part of this problem relies upon the fact that (U_1, U_2) is a disjoint pair, which is not explicitly stated in the problem. For why this is the case, see my note in Problem 13 of Chapter 6.

Suppose $\mathcal{S}$ is a Rosser system. Thus for any two recursively enumerable sets A and B, there is a number h that separates $A - B$ from $B - A$. Consider any completely effectively inseparable pair of recursively enumerable sets (we know that at least one exists, namely the Kleene pair (K_1, K_2)). Now, because K_1 and K_2 are disjoint, $K_1 - K_2 = K_1$ and $K_2 - K_1 = K_2$. Thus, because $\mathcal{S}$ is a Rosser system, K_1 is separable from K_2 in $\mathcal{S}$ by some number h. Thus h represents some superset S_1 of K_1 and contra-represents some superset S_2 of K_2. Clearly, S_1 and S_2 must be disjoint. For if they shared some number n, we would have $h(n) \in P$ and $h(n) \in R$, which is impossible. This means that (S_1, S_2) is also completely effectively inseparable. For consider any supersets ω_i and ω_j of S_1 and S_2 respectively. Now, ω_i and ω_j are also supersets of K_1 and K_2 respectively. Thus there is some recursive function $\zeta(x, y)$ (by the complete effective inseparability of (K_1, K_2)) such that $\zeta(i, j) \in \omega_i$ iff $\zeta(i, j) \in \omega_j$. Hence $\zeta(x, y)$ is a completely effective inseparability function for (S_1, S_2), meaning that pair is completely effectively inseparable. Now, by Theorem 9 of Chapter 6, (S_1, S_2) is also doubly universal. Hence, because h exactly separates (S_1, S_2), we have shown that there is an exactly separable doubly universal pair in $\mathcal{S}$.

Now suppose that some doubly universal pair is exactly separable in $\mathcal{S}$ and that all recursive functions of one argument are admissible in $\mathcal{S}$. Thus for every recursive function $f(x)$ and for every number h, there is a number k such that for every n the number $k(n)$ is equivalent to $h(f(n))$. Now consider some disjoint pair (A, B) of recursively enumerable sets. Now, consider the doubly universal pair (U_1, U_2) that was assumed exactly separable in $\mathcal{S}$. Thus, for some

recursive function $f(x)$ we have, for all x:
(1) $x \in A$ iff $f(x) \in U_1$;
(2) $x \in B$ iff $f(x) \in U_2$;
Now, we know that $f(x)$ is admissible in $\mathcal{S}$ (because it only has one argument). Consider the h that exactly separates (U_1, U_2). Thus there is some number k such that for every n the number $k(n)$ is equivalent to $h(f(n))$. Hence $k(n) \in P$ iff $h(f(n)) \in P$ iff $f(n) \in U_1$ iff $n \in A$, and $k(n) \in R$ iff $h(f(n)) \in R$ iff $f(n) \in U_2$ iff $n \in B$, thus showing that (A, B) is exactly separable by the number k.

Problem 24

Suppose that $\mathcal{S}$ is an effective Rosser system. Consider some recursively enumerable relations $R_1(x, y)$ and $R_2(x, y)$. Let $\Pi(x, y)$ be a Rosser function in $\mathcal{S}$. Thus $R_1(x, \Pi(x, y))$ and $R_2(x, \Pi(x, y))$ are both recursively enumerable. By the weak double recursion theorem, there are numbers a and b such that:
(1) $x \in \omega_a$ iff $R_1(x, \Pi(a, b))$;
(2) $x \in \omega_b$ iff $R_2(x, \Pi(a, b))$.
Now suppose that $R_1(n, \Pi(a, b)) \wedge \sim R_2(n, \Pi(a, b))$ holds. Thus $x \in \omega_a$ and $x \notin \omega_b$ by (1) and (2), respectively, that is, $x \in \omega_a - \omega_b$. Then, by the definition of a Rosser function $\Pi(x, y)$, $\Pi(a, b)(n) \in P$ (because $\Pi(a, b)$ represents some superset of $(\omega_a - \omega_b)$). On the other hand, suppose that $R_2(n, \Pi(a, b)) \wedge \sim R_1(n, \Pi(a, b))$ holds. Thus $x \in \omega_b$ and $x \notin \omega_a$ by (2) and (1), respectively, that is, $x \in \omega_b - \omega_a$. Then, by the definition of the Rosser function, $\Pi(a, b)(n) \in R$ (as $\Pi(a, b)$ contra-represents some superset of $\omega_b - \omega_a$). Thus we have shown:
(1′) $R_1(n, h) \wedge \sim R_2(n, h)$ implies $h(n) \in P$;
(2′) $R_2(n, h) \wedge \sim R_1(n, h)$ implies $h(n) \in R$;
when taking $\Pi(a, b)$ for h.

Problem 25

Suppose that $\mathcal{S}$ is an effective Rosser system. Consider any two recursively enumerable disjoint sets A and B. Recall that $y(x) = r(y, x)$, a recursive function. Let $R_1(x, y)$ be the relation $x \in A \vee y(x) \in R$ and $R_2(x, y)$ be the relation $x \in B \vee y(x) \in P$, which are both recursively enumerable. By Proposition 3, there is a number h such that for all n:
(1) $R_1(n, h) \wedge \sim R_2(n, h)$ implies $h(n) \in P$;
(2) $R_2(n, h) \wedge \sim R_1(n, h)$ implies $h(n) \in R$;
that is:
(1′) $(n \in A \vee h(n) \in R) \wedge \sim (n \in B \vee h(n) \in P)$ implies $h(n) \in P$;

(2′) $(n \in B \vee h(n) \in P) \wedge \sim(n \in A \vee h(n) \in R)$ implies $h(n) \in R$.
We must show that h exactly separates A from B, that is, $n \in A$ iff $h(n) \in P$ and $n \in B$ iff $h(n) \in R$.

First, suppose $h(n) \in P$. Thus, by (2′), $\sim((n \in B \vee h(n) \in P) \wedge \sim(n \in A \vee h(n) \in R))$ holds. $\sim(n \in B \vee h(n) \in P)$ cannot hold, or else $h(n) \notin P$, a contradiction. Thus $n \in A \vee h(n) \in R$ holds, and because $h(n) \in R$ cannot hold, we have $h(n) \in A$. On the other hand, suppose that $h(n) \in R$. Thus, by (1′), $\sim((n \in A \vee h(n) \in R) \wedge \sim(n \in B \vee h(n) \in P))$ holds. It cannot be that $\sim(n \in A \vee h(n) \in R)$ holds, or else $h(n) \notin R$, a contradiction. Thus $n \in B \vee h(n) \in P$ holds. But because $h(n) \in P$ cannot hold, $n \in B$.

Now we show the converses. First suppose that $n \in A$. Thus $n \in A \vee h(n) \in R$ holds, and $h(n) \notin R$, or else $n \in B$ by the above paragraph (and A and B are disjoint). Now, if $h(n) \notin P$, we would have $h(n)$ being undecidable. But because $n \notin B$ and $h(n) \notin P$, we would have $\sim(n \in B \vee h(n) \in P)$ holding. Thus $(n \in A \vee h(n) \in R) \wedge \sim(n \in B \vee h(n) \in P)$ holds, meaning $h(n) \in P$, a contradiction. Hence $h(n)$ cannot be undecidable, and consequently $h(n) \in P$. A similar argument proves that if $n \in B$, then $h(n) \in R$.

Thus we have shown that $n \in A$ iff $h(n) \in P$ and $n \in B$ iff $h(n) \in R$, meaning h exactly separates A from B. Thus we have shown, for any two disjoint recursively enumerable sets A and B, there is some number h that exactly separates A from B, meaning, by definition, $\mathcal{S}$ is an exact Rosser system.

Problem 26

Consider any effective Rosser system and any two recursive functions $f_1(x)$ and $f_2(x)$. Hence the relations $y(x) \in \omega_{f_1(x)}$ and $y(x) \in \omega_{f_2(x)}$ are recursively enumerable. By Proposition 3, there is a number h such that for any n:
(1) $(h(n) \in \omega_{f_1(n)}) \wedge \sim(h(n) \in \omega_{f_2(n)})$ implies $h(n) \in P$;
(2) $(h(n) \in \omega_{f_2(n)}) \wedge \sim(h(n) \in \omega_{f_1(n)})$ implies $h(n) \in R$.
Now consider any n for which $\omega_{f_1(n)}$ is disjoint from $\omega_{f_2(n)}$. This means that $h(n) \in \omega_{f_1(n)}$ and $h(n) \in \omega_{f_2(n)}$ imply $(h(n) \in \omega_{f_1(n)}) \wedge \sim(h(n) \in \omega_{f_2(n)})$ and $(h(n) \in \omega_{f_2(n)}) \wedge \sim(h(n) \in \omega_{f_1(n)})$, respectively. Thus, by (1) and (2):
(1′) $h(n) \in \omega_{f_1(n)}$ implies $h(n) \in P$;
(2′) $h(n) \in \omega_{f_2(n)}$ implies $h(n) \in R$.
for all n such that $\omega_{f_1(n)}$ and $\omega_{f_2(n)}$ are disjoint.

Note: For future reference, here is why $y(x) \in \omega_{f_1(x)}$ and $y(x) \in \omega_{f_2(x)}$ are recursively enumerable. $y(x) \in \omega_{f_1(x)}$ is equivalent to

$$\exists x_1 \exists x_2 (y(x) = x_1 \wedge f_1(x) = x_2 \wedge r(x_2, x_1) \in T_0)$$

which is recursively enumerable. Similarly, $y(x) \in \omega_{f_2(x)}$ is equivalent to

$$\exists x_1 \exists x_2 (y(x) = x_1 \wedge f_2(x) = x_2 \wedge r(x_2, x_1) \in T_0)$$

Problem 27

Consider some effective Rosser system, and any recursively enumerable sequence of extensions of $\mathcal{S}$. Thus the relations $x \in P_y$ and $x \in R_y$ are both recursively enumerable. By the iteration theorem, there are recursive functions $f_1(x)$ and $f_2(x)$ such that $\omega_{f_1(n)} = \{x : x \in R_n\}$ and $\omega_{f_2(n)} = \{x : x \in P_n\}$. That is, $\omega_{f_1(n)} = R_n$ and $\omega_{f_2(n)} = P_n$. Then, by the Rosser fixed point property, there is a number h such that for all n (because $\omega_{f_1(n)}$, that is R_n, is always disjoint from $\omega_{f_2(n)}$, that is P_n):
(1) $h(n) \in \omega_{f_1(n)}$ implies $h(n) \in P$;
(2) $h(n) \in \omega_{f_2(n)}$ implies $h(n) \in R$.
Now consider any n, and suppose that $h(n) \in P_n$, meaning $h(n) \in \omega_{f_2(n)}$, and thus, by (2), $h(n) \in R$. Recall that, because $\mathcal{S}_n$ is an extension of $\mathcal{S}$, $P \subseteq P_n$ and $R \subseteq R_n$. Thus $h(n) \in R_n$, a contradiction. Hence $h(n) \notin P_n$. Similar reasoning can be used to show that $h(n) \notin R_n$. Thus we have shown, for every n, $h(n) \notin P_n \cup R_n$, which proves that $\mathcal{S}$ is uniformly incompletable.